LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: INT396	Unit: 4
Course Title: UNSUPERVISED LEARNING	Unit Title: Dimensionality Reduction and Representation Learning
Course Outcome: CO4	Unit Mapped RBT: L5 (Evaluate)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Evaluate the decision to replace 200 correlated sensor channels with 15 principal components before a clustering is run. (10 Marks)

(a) Assess what is gained by the reduction and what is placed at risk by it.

(b) Recommend the checks that should be reported alongside the clustering, and justify each one.

[L5 - Evaluate]

Q2. Assess the claim that a t-SNE plot showing five tidy islands proves the customer base divides into five segments. (10 Marks)

(a) Judge what a t-SNE layout does and does not preserve about the original data.

(b) Recommend the evidence you would require before accepting the claim.

[L5 - Evaluate]

Q3. Justify a choice between feature selection and feature extraction for a model whose reduced inputs must be explained to a regulator. (10 Marks)

(a) Assess what each of the two approaches produces from the original columns.

(b) Justify your choice for this setting, and state the cost that the choice carries.

[L5 - Evaluate]

Q4. Critique the following statement taken from a project report: 'We kept enough components to reach 95 per cent of the variance, so nothing of value was lost.' (10 Marks)

(a) Assess what the variance figure does and does not establish.

(b) Recommend what the report should have presented instead.

[L5 - Evaluate]

Q5. Evaluate the use of an autoencoder in place of Principal Component Analysis on a dataset of 2,000 records described by 60 features. (10 Marks)

(a) Assess what an autoencoder is able to capture that Principal Component Analysis cannot.

(b) Judge whether the exchange is worthwhile at this size of dataset, and give your reasons.

[L5 - Evaluate]

Q6. Determine how many principal components should be retained for a given dataset, and defend the decision. (10 Marks)

(a) Assess the common rules for deciding, naming a weakness of each.

(b) Recommend one of them for a dataset whose scree plot has no clear bend, and justify the recommendation.

[L5 - Evaluate]

Q7. Assess the effect of the curse of dimensionality on any method that relies on distances. (10 Marks)

(a) Judge what happens to the distances between records as the number of features grows.

(b) Recommend two responses to the problem and assess the cost that each one carries.

[L5 - Evaluate]

Q8. Judge whether Principal Component Analysis should be applied to a table whose columns are recorded in different units without any preparation. (10 Marks)

(a) Assess what variance means when two columns are measured in different units.

(b) Recommend the correct procedure and justify why it is necessary.

[L5 - Evaluate]

Q9. Critique a two-dimensional embedding that has been placed in a report with no mention of the method used or its settings. (10 Marks)

(a) Assess what a reader of that report is unable to check.

(b) Recommend the minimum information that should accompany such a plot.

[L5 - Evaluate]

Q10. Evaluate dimensionality reduction as a step in an unsupervised workflow. (10 Marks)

(a) Assess three benefits that it can deliver to the analysis that follows.

(b) Judge the circumstances in which it should not be applied at all, and give the reasons.

[L5 - Evaluate]

----- End of Sample Question Set -----